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# FIND-S Algorithm Implementation
# Google Colab Compatible
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import pandas as pd

# Step 1: Create Training Dataset
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]

columns = ['Sky', 'AirTemp', 'Humidity', 'Wind',
           'Water', 'Forecast', 'EnjoySport']

df = pd.DataFrame(data, columns=columns)

print("Training Dataset:\n")
print(df)

# Step 2: Separate Features and Target
concepts = df.iloc[:, :-1].values
target = df.iloc[:, -1].values

# Step 3: Implement FIND-S Algorithm
def find_s(concepts, target):
    # Initialize with the first positive example
    hypothesis = None

    for i in range(len(target)):
        if target[i] == "Yes":
            hypothesis = concepts[i].copy()
            break

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# Generalize hypothesis with remaining positive examples
for i in range(len(concepts)):
    if target[i] == "Yes":
        for j in range(len(hypothesis)):
            if hypothesis[j] != concepts[i][j]:
                hypothesis[j] = '?'

return hypothesis

# Step 4: Find the Most Specific Hypothesis
final_hypothesis = find_s(concepts, target)

print("\nMost Specific Hypothesis:")
print(final_hypothesis)

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Training Dataset:

	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
0	Sunny	Warm	Normal	Strong	Warm	Same	Yes
1	Sunny	Warm	High	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Cool	Change	Yes

Most Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' '?']

